

Solution to Exercise 09 — Analysis on Manifolds, 80416

June 18, 2026



Question 1

In each case given a manifold M and vector field F we will calculate the flux of F through M with respect to the unit normal pointing outward in two ways

1. Straight from definition is the surface integral

$$\int_M \langle F, N \rangle d\sigma$$

2. Using the Divergence Theorem

$$\int_M \langle F, N \rangle d\sigma = \int_{\partial M} \operatorname{div}(F) dx$$

Subquestion 1

$$M = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = z, z \leq 9\} \cup \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 9, z = 9\}$$

$$F(x, y, z) = (7xz, xz^2, x^2y^2)$$

Solution:

1. Straight from definition – we need to split in two parts M_1, M_2 so that $M = M_1 \cup M_2$ given by

$$M_1 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = z, z \leq 9\}$$

$$M_2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 9, z = 9\}$$

For M_1 , we take the parametrization $\Phi_1(r, \theta) = (r \cos(\theta), r \sin(\theta), r^2)^T$ for $r \in [0, 3]$ and $\theta \in [0, 2\pi]$.

Let

$$\Phi_1 = \begin{pmatrix} r \cos(\theta) \\ r \sin(\theta) \\ r^2 \end{pmatrix}$$

We compute

$$\frac{\partial \Phi_1}{\partial r} = \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 2r \end{pmatrix}, \quad \frac{\partial \Phi_1}{\partial \theta} = \begin{pmatrix} -r \sin(\theta) \\ r \cos(\theta) \\ 0 \end{pmatrix}$$

We need to compute their cross product

$$\frac{\partial \Phi_1}{\partial r} \times \frac{\partial \Phi_1}{\partial \theta} = \begin{vmatrix} i & j & k \\ \cos(\theta) & \sin(\theta) & 2r \\ -r \sin(\theta) & r \cos(\theta) & 0 \end{vmatrix}$$

Reminder, if

$$a \times b = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2 b_3 - a_3 b_2)i - (a_1 b_3 - a_3 b_1)j + (a_1 b_2 - a_2 b_1)k$$

So in our case

$$\frac{\partial \Phi_1}{\partial r} \times \frac{\partial \Phi_1}{\partial \theta} = \begin{pmatrix} -2r^2 \cos(\theta) \\ -2r^2 \sin(\theta) \\ r \end{pmatrix}$$

If we look at the z -component, since $0 \leq r \leq 3$ we have a positive component which means this vector points upward but from the bottom which means inward so we need to multiple by -1 (to turn down and away) hence

$$N_1 d\sigma = \begin{pmatrix} 2r^2 \cos(\theta) \\ 2r^2 \sin(\theta) \\ -r \end{pmatrix} dr d\theta$$

Now

$$F(\Phi_1(r, \theta)) = \begin{pmatrix} 7r^3 \cos(\theta) \\ r^5 \cos(\theta) \\ r^4 \cos^2(\theta) \sin^2(\theta) \end{pmatrix}$$

Hence

$$\langle F(\Phi_1(r, \theta)), N_1 d\sigma \rangle = 14r^5 \cos^2(\theta) + 2r^7 \cos(\theta) \sin(\theta) - r^5 \cos^2(\theta) \sin^2(\theta)$$

Finally,

$$\int_0^{2\pi} \int_0^3 14r^5 \cos^2(\theta) + 2r^7 \cos(\theta) \sin(\theta) - r^5 \cos^2(\theta) \sin^2(\theta) d\theta dr$$

Which looks scary but we know that

$$\int_0^{2\pi} 2r^7 \cos(\theta) \sin(\theta) d\theta = 0$$

If we use the u-substitution and

$$\int_0^{2\pi} \cos^2(\theta) d\theta = \pi \implies \int_0^{2\pi} 14r^5 \cos^2(\theta) d\theta = 14\pi r^5$$

From the identity $\cos(\theta) \sin(\theta) = \frac{1}{2} \sin(2\theta)$ and from $\int_0^{2\pi} \sin^2(2\theta) d\theta = \pi$ one can obtain that

$$\int_0^{2\pi} -r^5 \cos^2(\theta) \sin^2(\theta) d\theta = -\frac{\pi}{4} r^5$$

So we have over M_1

$$\int_0^3 \frac{55}{4} \pi r^5 dr = \frac{13365}{8} \pi$$

For M_2 , we have the surface $z = 9$ over the disk $x^2 + y^2 \leq 9$ and the outward normal pointing straight up is $N_2 = (0, 0, 1)^T$ and $d\sigma = dx dy$ and $\langle F, N_2 \rangle = (7xz, xz^2, x^2y^2) \cdot (0, 0, 1) = x^2y^2$ and using polar coordinates we can compute

$$\iint_{M_2} x^2y^2 dx dy = \int_0^{2\pi} \int_0^3 (r \cos(\theta))^2 (r \sin(\theta))^2 r dr d\theta = \int_0^{2\pi} \int_0^3 r^5 dr \int_0^{2\pi} \cos^2(\theta) \sin^2(\theta) d\theta$$

obviously,

$$\int_0^3 r^5 dr = \left[\frac{r^6}{6} \right]_{r=0}^{r=3} = 121.5$$

But the other integral is more complicated so we use the identities

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}, \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

Will provide us the following

$$\int_0^{2\pi} \cos^2(\theta) \sin^2(\theta) d\theta = \int_0^{2\pi} \frac{(1 - \cos(2\theta))(1 + \cos(2\theta))}{4} d\theta = \int_0^{2\pi} \frac{1 - \cos^2(2\theta)}{4} d\theta = \frac{1}{4} \left(\int_0^{2\pi} 1 d\theta - \int_0^{2\pi} \cos^2(2\theta) d\theta \right)$$

The first integral is obvious π , for the other one we use the change of variable with $u = 2\theta$, $\theta = \frac{u}{2}$ and $d\theta = \frac{1}{2} du$. Without the integral limits we get that

$$\int \cos^2(2\theta) d\theta = \frac{1}{2} \int \cos^2(u) du = \frac{1}{2} \int \frac{\cos(2u) + 1}{2} du = \frac{1}{4} \left(\int \cos(2u) du + \int 1 du \right) = \frac{1}{4} \int \cos(2u) du + \frac{u}{4}$$

With another change of variable with $v = 2u$, $u = \frac{v}{2}$ and $du = \frac{1}{2} dv$ we get that

$$\int \cos(2u) du = \frac{1}{2} \int \cos(v) dv = \frac{\sin(v)}{2} = \frac{\sin(2u)}{2} = \frac{\sin(4\theta)}{2}$$

Finally if we substitute everything back

$$\int_0^{2\pi} \cos^2(\theta) \sin^2(\theta) d\theta = \frac{1}{4} \left(\int_0^{2\pi} 1 d\theta - \int_0^{2\pi} \cos^2(\theta) d\theta \right) = \frac{1}{4} \left(2\pi - \left[\frac{\sin(4\theta)}{8} + \frac{\theta}{2} \right]_{\theta=0}^{\theta=2\pi} \right) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

In total

$$\iint_{M_2} x^2 y^2 dx dy = 121.5 \cdot \frac{\pi}{4} = \frac{243}{8} \pi$$

We sum everything up and get that

$$\text{Total Flux} = \frac{13365}{8} \pi + \frac{243}{8} \pi = 1701 \pi$$

2. Using the Divergence Theorem – we start by computing the divergence

$$\nabla \cdot F = \frac{\partial}{\partial x}(7xz) + \frac{\partial}{\partial y}(xz^2) + \frac{\partial}{\partial z}(x^2 + y^2) = 7z + 0 + 0 = 7z$$

Our region is bounded below by $z = x^2 + y^2$ and above by that constant plane $z = 9$ so it is natural to choose cylindrical coordinates with

$$x = r \cos(\theta), \quad y = r \sin(\theta), \quad z = z, \quad dV = r dz dr d\theta$$

Where $\theta \in [0, 2\pi]$, $0 \leq r \leq 3$ (since $x^2 + y^2 = 9 \implies r^2 = 9$) and $r^2 \leq z \leq 9$ hence we can compute

$$\begin{aligned} \int_M \langle F, N \rangle d\sigma &= \int_{\partial M} \text{div}(F) dx \\ &= \int_0^{2\pi} \int_0^3 \int_{r^2}^9 7zr dz dr d\theta \\ &= \int_0^{2\pi} \int_0^3 \left[\frac{7z^2 r}{2} \right]_{z=r^2}^{z=9} dr d\theta \\ &= \int_0^{2\pi} \int_0^3 \frac{567}{2} r - \frac{7}{2} r^5 dr d\theta \\ &= \int_0^{2\pi} \left[\frac{567}{4} r^2 - \frac{7}{12} r^6 \right]_{r=0}^{r=3} d\theta \\ &= \int_0^{2\pi} \frac{1701}{2} d\theta \\ &= 1701\pi \end{aligned}$$

□

Subquestion 2

$$M = \partial C$$

$$C = \{(x, y, z) \in \mathbb{R}^3 \mid y^2 + z^2 < 9, 0 < x < 2\}$$

$$F(x, y, z) = (x^2, -y, z)$$

Solution:

1. Straight from definition – First we understand the C is a bounded slice of a cylinder over the yz plane with radius 3 with the constraint that $0 < x < 2$ so its boundary, $M = \partial C$, contains the following smooth surfaces

1. M_1 , This is like the confetti rounds for a Soka with the normal from the x -axis is $\hat{n}_1 = (0, \frac{y}{3}, \frac{z}{3})$ and given by

$$M_1 := \{(x, y, z) \in \mathbb{R}^3 \mid y^2 + z^2 = 9, 0 \leq x \leq 2\}$$

The normal was calculated as the following: for $G(x, y, z) = y^2 + z^2$ and $G = 9$ so

$$\nabla G = (0, 2y, 2z), \quad \|\nabla G\| = \sqrt{0^2 + (2y)^2 + (2z)^2} = 2\sqrt{y^2 + z^2} \underset{y^2+z^2=9}{=} 2\sqrt{9} = 6$$

$$\Rightarrow \hat{n}_1 = \frac{\nabla G}{\|\nabla G\|} = \frac{(0, 2y, 2z)}{6} = \left(0, \frac{2y}{6}, \frac{2z}{6}\right) = \left(0, \frac{y}{3}, \frac{z}{3}\right)$$

2. M_2 , The right flat disk with outward normal $\hat{n}_2 = (1, 0, 0)$ and given by

$$M_2 := \{(x, y, z) \in \mathbb{R}^3 \mid y^2 + z^2 \leq 9, x = 0\}$$

3. M_3 , The left flat disk with outward normal $\hat{n}_3 = (-1, 0, 0)$ and given by

$$M_3 := \{(x, y, z) \in \mathbb{R}^3 \mid y^2 + z^2 \leq 9, x = 2\}$$

Now for each one of them we will compute

$$\int_{M_i} \langle F, \hat{n}_i \rangle d\sigma$$

1. For M_1 – We first understand that on M_1 we can slide forward and backward along the x -axis and it doesn't change our y or z position so we must integrate by dx .

Furthermore, if we look at $y^2 + z^2 = 9$ it is natural to choose the mapping $y = 3 \cos(\theta)$, $z = 3 \sin(\theta)$ (since we are in a circle of radius 3) and to trace the entire circle $\theta \in [0, 2\pi]$ so in our case we have

$$\psi(x, \theta) = (x, 3 \cos(\theta), 3 \sin(\theta))$$

ψ is well defined so by the change of variable theorem we have $d\sigma = 3 dx d\theta$ We start by computing

$$\langle F, \hat{n}_1 \rangle = (x^2, -y, z) \cdot \left(0, \frac{y}{3}, \frac{z}{3}\right) = \frac{-y^2 + z^2}{3}$$

Hence

$$\begin{aligned} \int_{M_1} \langle F, \hat{n}_1 \rangle d\sigma &= \int_0^{2\pi} \int_0^2 \frac{-9 \cos^2(\theta) + 9 \sin^2(\theta)}{3} 3 dx d\theta = 9 \int_0^2 dx \int_0^{2\pi} \sin^2(\theta) - \cos^2(\theta) d\theta = 9 \int_0^2 dx \int_0^{2\pi} -\cos(2\theta) d\theta \\ &= 9[x]_{x=0}^{x=2} \left[-\frac{\sin(2\theta)}{2} \right]_{\theta=0}^{\theta=2\pi} = 18 \cdot 0 = 0 \end{aligned}$$

2. For M_2 – Substituting $x = 2$ into F gives us $F(2, y, z) = (4, -y, z)$ and $\langle F, \hat{n}_2 \rangle = 4$ hence

$$\int_{M_2} \langle F, \hat{n}_2 \rangle d\sigma = \int_{M_2} 4 d\sigma = 4 \times \text{Area}(M_2) = 4 \cdot (\pi \cdot 3^2) = 36\pi$$

3. For M_3 – Substituting $x = 0$ into F gives us $F(0, y, z) = (0, -y, z)$ and $\langle F, \hat{n}_3 \rangle = 0$ hence

$$\int_{M_3} \langle F, \hat{n}_3 \rangle d\sigma = \int_{M_3} 0 d\sigma = 0$$

Hence the total flux through all three surfaces is $0 + 36\pi + 0 = 36\pi$

2. Using the Divergence Theorem – we start by computing the divergence

$$\nabla \cdot F = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(-y) + \frac{\partial}{\partial z}(z) = 2x + -1 + 1 = 2x$$

We notice that we don't need to use any other coordinates system since we need to compute

$$\int_C 2x \, dV = \int_0^2 \left(\int \int_{y^2+z^2 \leq 9} 2x \, dy \, dz \right) dx$$

But the inner integral is the just double integral with respect to y and z that represents the area of the disk D with radius 3 multiplied by the constant $2x$ so it just

$$\int_C 2x \, dV = \int_0^2 2x \cdot \text{Area}(D) \, dx = \int_0^2 2x(9\pi) \, dx = 18\pi \int_0^2 x \, dx = 18\pi \left[\frac{x^2}{2} \right]_0^2 = 18\pi \left(\frac{4}{2} - 0 \right) = 36\pi$$

□

Question 2

Let $0 < r < R$ and consider the set $T = \{(x, y, z) \in \mathbb{R}^3 \mid (\sqrt{x^2 + y^2} - R)^2 + z^2 = r^2\}$.

This is a 2-dimensional manifold which called a torus.

Subquestion 1

For every $z_0 \in \mathbb{R}$ we will find the set $T \cap \{z = z_0\}$ and draw a sketch of the torus.

Solution: We substitute $z = z_0$ into the definition so we will have

$$(\sqrt{x^2 + y^2} - R)^2 + z_0^2 = r^2 \Leftrightarrow (\sqrt{x^2 + y^2} - R)^2 = r^2 - z_0^2$$

We split into cases

1. $|z_0| > r$ hence $r^2 - z_0^2 < 0$ and the intersection will be only the $\emptyset$
2. $|z_0| = r$ so $r^2 - z_0^2 = 0$ so the equation will become

$$\sqrt{x^2 + y^2} - R = 0 \Leftrightarrow x^2 + y^2 = R^2$$

So we have a circle of radius R centered at the z -axis on the plane $\pm r$ which means the intersection is the set $\{(x, y, z_0) \in \mathbb{R}^3 \mid x^2 + y^2 = R^2\}$

3. $0 \leq |z_0| < r$ hence $r^2 - z_0^2 > 0$ so we can take a square root and achieve that

$$\sqrt{x^2 + y^2} - R = \pm \sqrt{r^2 - z_0^2} \Leftrightarrow \sqrt{x^2 + y^2} = R \pm \sqrt{r^2 - z_0^2}$$

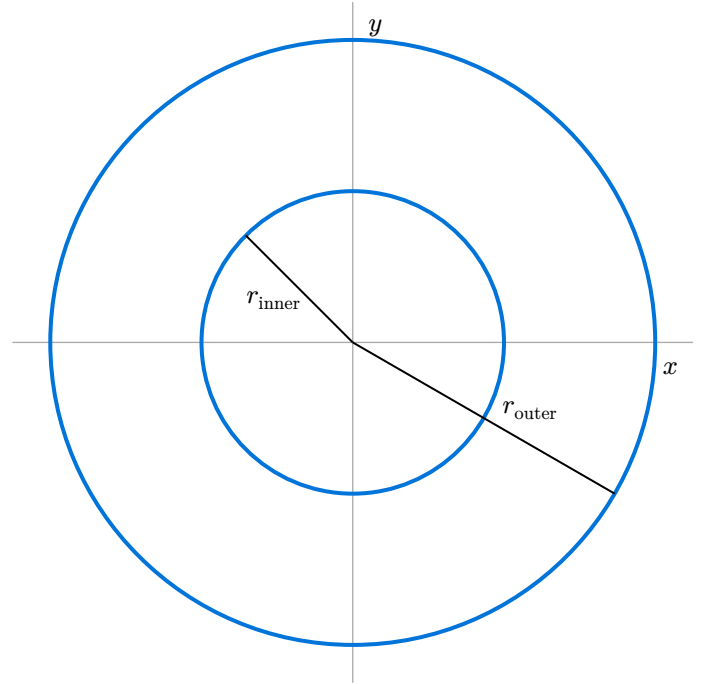
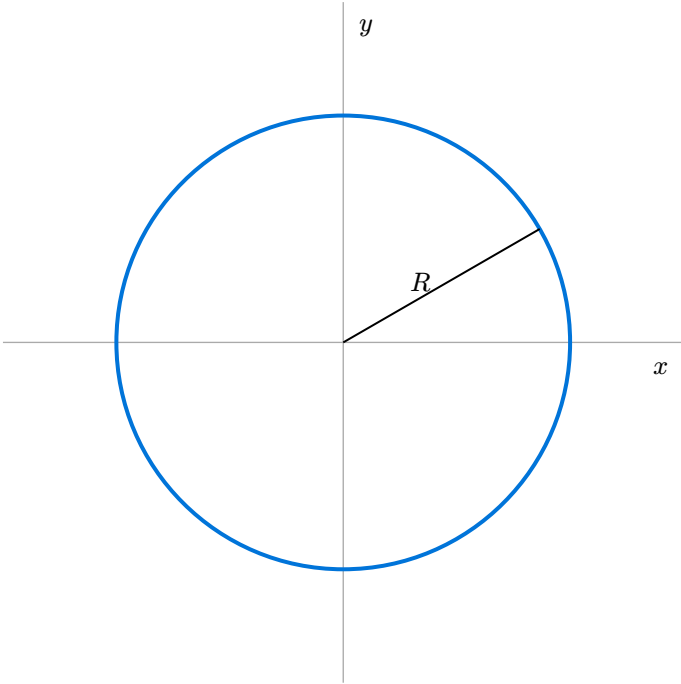
We know that $r < R$ so the term $\sqrt{r^2 - z_0^2} < R$ hence the right side is positive which gives as two circle equations

$$x^2 + y^2 = \left(R + \sqrt{r^2 - z_0^2}\right)^2$$

$$x^2 + y^2 = \left(R - \sqrt{r^2 - z_0^2}\right)^2$$

Which gives circles centered on the z -axis with the above radiuses.

I don't really know how to draw but I guess it is obvious from below.



□

Subquestion 2

We will calculate the flux of $F(x, y, z) = (1, ze^x, -ye^x)$ through $T \cap \{(x, y, z) \in \mathbb{R}^3 \mid z > 0\}$ with respect to the unit normal pointing upward.

Solution: From the previous subquestion we define

$$V := \left\{ (x, y, z) \in \mathbb{R}^3 \mid \left(\sqrt{x^2 + y^2} - R \right)^2 + z^2 \leq r^2, z \geq 0 \right\}$$

We need to understand ∂V but the bad drawing above helps us! The boundary contains the top “curs”, M , where $z \geq 0$ and the outward unit normal $\hat{n}_M$ and the flat bottom, D , is where $z = 0$ and from the previous subquestion we know this region is bounded by the circles of radius $R \pm r$ and in this case the outward unit normal here will point downward which means $\hat{n}_D = (0, 0, -1)$.

Now, computing $\text{div}(F)$

$$\nabla \cdot F = \frac{\partial}{\partial x} 1 + \frac{\partial}{\partial y} ze^x + \frac{\partial}{\partial z} -ye^x = 0 + 0 + 0 = 0$$

By the Divergence Theorem we have

$$0 = \iiint_V \nabla \cdot F \, dV = \int_M \langle F, \hat{n}_M \rangle \, d\sigma + \int_D \langle F, \hat{n}_D \rangle \, d\sigma \implies \int_M \langle F, \hat{n}_M \rangle \, d\sigma = - \int_D \langle F, \hat{n}_D \rangle \, d\sigma$$

Since $\hat{n}_D = (0, 0, -1)$ the right side can be computed easily (we remember that $z = 0$ on D)

$$\langle F, \hat{n}_D \rangle = (1, 0, -ye^x) \cdot (0, 0, -1) = ye^x$$

So the integral on the right is just

$$\int_D \langle F, \hat{n}_D \rangle \, d\sigma = \iint_D ye^x \, dx \, dy$$

First, we need to remember that in this domain $(R - r)^2 \leq x^2 + y^2 \leq (R + r)^2$ so the top half of the shape is a mirror image of the bottom half (over the x -axis) so if we fix a x , we can see that

$$f(x, y) = ye^x \implies f(x, -y) = -ye^x = -f(x, y)$$

So the integrand is an odd function with respect to y and integrating an odd function over symmetric domain the integral is exactly zero so if we apply Fubini's Theorem to swap the order of the integration we will get that

$$\iint_D \langle F, \hat{n}_D \rangle \, d\sigma = \iint_D ye^x \, dx \, dy = \iint_D ye^x \, dy \, dx = 0$$

Hence we also have

$$\int_M \langle F, \hat{n}_M \rangle \, d\sigma = 0$$

Therefore the wanted flux is zero. □

Question 3

Subquestion 1

Let $f : B_r(p) \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function. We will show that

$$\int_{B_r(p)} f \, dx = \int_0^r \int_{\partial B_t(p)} f \, d\sigma \, dt$$

Proof: It is suffice to show for $p = 0$ since we can always use a translation to move our point to the origin which is a continuous map. In $\mathbb{R}^3$ it is natural to use spherical coordinates mapping $\Psi : V = (0, r) \times (0, \pi) \times (0, 2\pi) \rightarrow \mathbb{R}^3$ given by

$$\Psi(t, \varphi, \theta) = \begin{pmatrix} t \sin(\varphi) \cos(\theta) \\ t \sin(\varphi) \sin(\theta) \\ t \cos(\varphi) \end{pmatrix}$$

We notice that $\Psi(V)$ covers the entire $B_r(0)$ except for a set of measure zero (the origin and a boundary half-plane) so by the Change of Variables Theorem

$$\int_{B_r(0)} f(x) \, dx = \int_V f(\Psi(t, \varphi, \theta)) |\det D\Psi(t, \varphi, \theta)| \, dt \, d\varphi \, d\theta$$

Where $\det D\Psi(t, \varphi, \theta) = t^2 \sin \varphi$ and since $\varphi \in (0, \pi)$ we have $\sin \varphi > 0$ which means we can drop the absolute values, so we have

$$\int_{B_r(0)} f(x) \, dx = \int_0^r \int_0^\pi \int_0^{2\pi} f(\Psi(t, \varphi, \theta)) t^2 \sin \varphi \, d\theta \, d\varphi \, dt \stackrel{\text{Fubini}}{=} \int_0^r \left(\int_0^\pi \int_0^{2\pi} f(\Psi(t, \varphi, \theta)) t^2 \sin \varphi \, d\theta \, d\varphi \right) dt$$

On the other hand, we fix t and look at $\partial B_t(0)$, this is a 2-dimensional manifold (the boundary of a manifold is a manifold without boundary) we take parametrization using the mapping $\gamma : (0, \pi) \times (0, 2\pi) \rightarrow \mathbb{R}^3$ defined by

$$\gamma(\varphi, \theta) = \Psi(t, \varphi, \theta) = \begin{pmatrix} t \sin \varphi \cos \theta \\ t \sin \varphi \sin \theta \\ t \cos \varphi \end{pmatrix}$$

We know that

$$d\sigma = \left\| \frac{\partial \gamma}{\partial \varphi} \times \frac{\partial \gamma}{\partial \theta} \right\| d\varphi \, d\theta$$

When

$$\frac{\partial \gamma}{\partial \theta} = \begin{pmatrix} t \cos \varphi \cos \theta \\ t \cos \varphi \sin \theta \\ -t \sin \varphi \end{pmatrix}, \quad \frac{\partial \gamma}{\partial \varphi} = \begin{pmatrix} -t \sin \varphi \sin \theta \\ t \sin \varphi \cos \theta \\ 0 \end{pmatrix}$$

Computing their cross product

$$\frac{\partial \gamma}{\partial \varphi} \times \frac{\partial \gamma}{\partial \theta} = t^2 \sin \varphi \begin{pmatrix} \sin \varphi \cos \theta \\ \sin \varphi \sin \theta \\ \cos \varphi \end{pmatrix} \implies \left\| \frac{\partial \gamma}{\partial \varphi} \times \frac{\partial \gamma}{\partial \theta} \right\| = t^2 \sin \varphi \implies d\sigma = t^2 \sin \varphi \, d\theta \, d\varphi$$

Hence

$$\int_0^\pi \int_0^{2\pi} f(\Psi(t, \varphi, \theta)) t^2 \sin \varphi \, d\theta \, d\varphi = \int_{\partial B_t(0)} f(x) \, d\sigma$$

But from the above

$$\int_{B_r(0)} f(x) \, dx = \int_0^r \left(\int_{\partial B_t(0)} f(x) \, d\sigma \right) dt$$

Which is exactly what we wanted to show. □

Subquestion 2

let $\Omega \subseteq \mathbb{R}^n$ be an open domain and let $u : \Omega \rightarrow \mathbb{R}$ be harmonic. prove that for every $p \in \Omega$ and $r > 0$ such that $B_r(p) \subseteq \Omega$ we have

$$\frac{1}{|B_r(p)|} \int_{B_r(p)} u(x) \, dx = u(p)$$

Proof: Using the change of variable to n -dimensional spherical coordinates we can write any point $x \in B_r(p)$ as $x = p + tw$ where $t \in [0, r]$ is the radial distance and $w \in S^{n-1}$ is a point on the unit sphere which corresponds to $\partial B_1(0)$ with $dx = t^{n-1} dt \, d\sigma_1(w)$ where $d\sigma_1$ is the surface area element on the unit sphere S^{n-1} hence

$$\int_{B_r(p)} f(x) \, dx = \int_0^r \int_{\partial B_1(0)} f(p + tw) t^{n-1} \, d\sigma_1(w) \, dt$$

For a fix t , the surface measure $d\sigma_t$ on $\partial B_t(p)$ is $t^{n-1} d\sigma_1$ hence

$$\int_{\partial B_t(p)} f(y) \, d\sigma_t(y) = \int_{\partial B_1(0)} f(p + tw) t^{n-1} \, d\sigma_1(w)$$

Which means that

$$\int_{B_r(p)} f(x) \, dx = \int_0^r \int_{\partial B_t(p)} f \, d\sigma \, dt$$

□

Subquestion 3

Let $\Omega \subseteq \mathbb{R}^n$ be an open domain and let $u : \Omega \rightarrow \mathbb{R}$ be harmonic. We will prove that for every $p \in \Omega$ and $r > 0$ such that $B_r(p) \subseteq \Omega$ we have

$$\frac{1}{|B_r(p)|} \int_{B_r(p)} u(x) \, dx = u(p)$$

Proof: From Gauss's mean value property we saw in the recitation we have for every $p \in \Omega$ and for every $t > 0$ such that $\overline{B_t(p)} \subseteq \Omega$ we have

$$u(p) = \frac{1}{|\partial B_t(0)|} \int_{\partial B_t(p)} u \, d\sigma$$

So if we rewrite it we will get

$$\int_{\partial B_t(p)} u \, d\sigma = u(p) |\partial B_t(0)|$$

From the previous subquestion

$$\int_{B_r(p)} u(x) \, dx = \int_0^r \int_{\partial B_t(p)} u \, d\sigma \, dt = \int_0^r u(p) |\partial B_t(0)| \, dt = u(p) \int_0^r |\partial B_t(0)| \, dt$$

But the integral on the right is just the integration of the surface areas of sphere from 0 to r which is just the volume of the solid ball $|B_r(0)|$ which means

$$\int_{B_r(p)} u(x) \, dx = u(p) |B_r(p)| \implies u(p) = \frac{1}{|B_r(p)|} \int_{B_r(p)} u(x) \, dx$$

□

Question 4

Let $\Omega \subseteq \mathbb{R}^n$ be a bounded domain, let $U \subseteq \mathbb{R}^n$ be an open neighborhood of $\overline{\Omega}$ and let $v : U \rightarrow \mathbb{R}$ be a function such that $\Delta v \geq 0$ (a subharmonic function). We will prove that

$$\max_{x \in \overline{\Omega}} v(x) = \max_{x \in \partial\Omega} v(x)$$

Proof: We start with proving the hint same as in the recitation, we define

$$f(r) = \frac{1}{|\partial B_r(0)|} \int_{\partial B_r(p)} v \, d\sigma \stackrel{\text{Change of Variable}}{=} \frac{1}{|\partial B_1(0)|} \int_{\partial B_1(0)} v(p + ry) \, d\sigma(y)$$

Differentiating under the integral sign gives us

$$\frac{d}{dr} f(r) = \frac{1}{|\partial B_r(0)|} \int_{\partial B_r(p)} \langle \nabla v, \hat{n} \rangle$$

From the Divergence Theorem

$$\frac{d}{dr} f(r) = \frac{1}{|\partial B_r(0)|} \int_{B_r(p)} \operatorname{div}(\nabla v) \, dx = \frac{1}{|\partial B_r(0)|} \int_{B_r(p)} \Delta v \, dx$$

Since v is subharmonic then $\Delta v \geq 0$ which means that $\frac{d}{dr} f(r) \geq 0$ which means it is a non-decreasing function of r and it is continuous so as $r \rightarrow 0$ we have $\lim_{r \rightarrow 0^+} f(r) = v(p)$ and since f is non-decreasing, $f(r) \geq v(p)$ for all $r > 0$ which implies

$$\frac{1}{|\partial B_r(0)|} \int_{\partial B_r(p)} v \, d\sigma \geq v(p) \implies \int_{\partial B_r(p)} v \, d\sigma \geq v(p) |\partial B_r(0)|$$

If we integrate over $t \in [0, r]$ using the result from question 3b

$$\int_{B_r(p)} v(x) \, dx = \int_0^r \int_{\partial B_t(p)} v \, d\sigma \, dt \geq \int_0^r v(p) |\partial B_t(0)| \, dt = v(p) |B_r(p)| \implies v(p) \leq \frac{1}{|B_r(p)|} \int_{B_r(p)} v(x) \, dx$$

Now, $\overline{\Omega}$ is a compact (since Ω is bounded) and v is continuous so v achieve maximum value M on $\overline{\Omega}$.

Suppose M is achieved at some interior point $p \in \Omega$ so that $v(p) = M$, we look at a ball $B_r(p) \subseteq \Omega$ and from the inequality above

$$M = v(p) \leq \frac{1}{|B_r(p)|} \int_{B_r(p)} v(x) \, dx$$

M is the maximum so in $\overline{\Omega}$ we know $v(x) \leq M$ for all $x \in B_r(p)$.

Since the average of a function that is strictly less than M on some positive measure set would be strictly less than M hence the only way the average over the ball can be at least M is if $v(x) = M$ everywhere in $B_r(p)$, which implies that the set

$$A := \{x \in \Omega \mid v(x) = M\}$$

is open but since v is continuous then A is also closed in Ω but in a connected domain the only subsets that are both open and closed are the entire set, Ω or $\emptyset$ and $p \in A$ hence $A = \Omega$ and from continuity, $v(x) = M$ on $\partial\Omega$ as well. $\square$